

What is the difference between "a subset" and "a subspace"?

Answer:

The core difference is that **"subset" is a general term**, while **"subspace" is a very specific and structured type of subset**.

In short: **every subspace is a subset, but not every subset is a subspace**.

For a subset to earn the special title of "subspace," it must be a well-behaved, self-contained vector space living inside a larger one (for example inside $\mathbf{R}^n$). In the course MA1513, we define a subspace (of $\mathbf{R}^n$) to be the linear span of a (finite) set of vectors (in $\mathbf{R}^n$).

To tell whether a (non-empty) subset W of a vector space V is a subspace of V , we can check whether W satisfies three specific conditions.

Here are the three conditions:

1. **It must contain the zero vector.** The zero vector $\mathbf{0}$ of the parent space V must be an element of the subset W . This is the most basic check. If the subset doesn't include the origin, it cannot be a subspace.
2. **It must be closed under vector addition.** If you take any two vectors $\mathbf{u}$ and $\mathbf{v}$ that are in the subset W , their sum, $\mathbf{u} + \mathbf{v}$, must also be in W . This ensures that the operation of addition doesn't "leave" the subset.
3. **It must be closed under scalar multiplication.** If you take any vector $\mathbf{u}$ that is in the subset W and multiply it by any scalar c (typically a real number), the resulting vector, $c\mathbf{u}$, must also be in W .

Conversely, if the subset W does not satisfy any one of the three conditions above, then W is not a subspace of V .

Scenario 1

Consider the vector space $V = \mathbf{R}^2$ (the entire 2D plane).

Let W be the set of all vectors lying on the line $y = 2x$. A vector that lies on this line is $(1, 2)$ (check that $x = 1, y = 2$ satisfy the equation of the line). So the subset W is given by the linear span of the vector $(1, 2)$, i.e. $W = \text{span}\{(1, 2)\}$. (Note that any scalar multiple of the vector $(1, 2)$ satisfies the equation $y = 2x$.) Therefore W is a subspace of V .

Let us check that W satisfies the three conditions of subspace:

1. The zero vector $(0, 0)$ is on the line, since $0 = 2 * 0$. So, W contains the zero vector.
2. If we take two vectors on the line, $\mathbf{u} = (x_1, 2x_1)$ and $\mathbf{v} = (x_2, 2x_2)$, their sum is $\mathbf{u} + \mathbf{v} = (x_1 + x_2, 2x_1 + 2x_2) = (x_1 + x_2, 2(x_1 + x_2))$. This new vector also fits the $y = 2x$ form, so it's on the line. The set W is closed under addition.
3. If we take a vector $\mathbf{u} = (x, 2x)$ and multiply it by a scalar c , we get $c\mathbf{u} = (cx, c(2x)) = (cx, 2(cx))$. This vector is also on the line. The set W is closed under scalar multiplication.

So W meets all three conditions of a subspace.

Scenario 2

Consider the vector space $V = \mathbf{R}^2$.

Let S be the subset of V consisting of all vectors lying on the line $y = 2x + 1$.

This line does not pass through the origin, because $0 \neq 2(0) + 1$. So the zero vector $(0, 0)$ does not belong to S . Since it fails one of the conditions of subspace, it is enough to conclude that the subset S is not a subspace of $\mathbf{R}^2$.

In fact, S does not satisfy the other two conditions as well. For example, if we take a vector $(1, 3)$ on this line (check that $x = 1, y = 3$ satisfy the equation $y = 2x + 1$), the scalar multiple $2(1, 3) = (2, 6)$ no longer lies on this line (check that $x = 2, y = 6$ do not satisfy the equation $y = 2x + 1$). Hence the subset S is not closed under scalar multiplication. It is also for this reason that S cannot be written as a linear span. For example, while $(1, 3)$ belongs to S , we have seen that $S \neq \text{span}\{(1, 3)\}$.

Scenario 3

Consider the vector space $V = \mathbf{R}^2$.

Let S be the set of V consisting of all vectors lying on the parabola curve $y = x^2$.

Note that this curve does pass through the origin, because $x = 0, y = 0$ satisfy the equation $y = x^2$. However, this is not sufficient to conclude whether the subset S is a subspace of $\mathbf{R}^2$ or not.

In fact, S does not satisfy the other two conditions.

1. If we take two specific vectors on the curve, $\mathbf{u} = (1, 1)$ and $\mathbf{v} = (2, 4)$, their sum is $\mathbf{u} + \mathbf{v} = (1+2, 1+4) = (3, 5)$. This new vector does not satisfy the $y = x^2$, so it's not on the curve. The set S is not closed under addition.
2. If we take the specific vector $\mathbf{u} = (1, 1)$ and multiply it by a scalar 2, we get $2\mathbf{u} = (2, 2)$. This new vector does not satisfy the $y = x^2$, so it's not on the curve. The set S is not closed under scalar multiplication.

Any one of the two violations above allows us to conclude that S is not a subspace of $\mathbf{R}^2$.

Scenario 4

Let W be the solution set of homogeneous system

$$\begin{aligned}x_1 + 2x_2 - x_3 + 3x_4 &= 0 \\2x_1 + 4x_2 + x_3 + 0x_4 &= 0\end{aligned}$$

As the system has 4 variables (x_1, x_2, x_3 , and x_4), this solution set W can be regarded as a subset of $\mathbf{R}^4$. For example, the trivial solution $x_1 = 0, x_2 = 0, x_3 = 0$ and $x_4 = 0$ gives the zero vector $(0, 0, 0, 0)$ in $\mathbf{R}^4$.

The augmented matrix for this system is: $\begin{bmatrix} 1 & 2 & -1 & 3 & 0 \\ 2 & 4 & 1 & 0 & 0 \end{bmatrix}$

After performing Gauss Jordan elimination, we get the RREF $\begin{bmatrix} 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & -2 & 0 \end{bmatrix}$.

The columns of variables x_2 and x_4 are non-pivot columns, so we set them as parameters $x_2 = s$ and $x_4 = t$.

After expressing x_1 and x_3 in terms of the parameters s and t , we get the solution

$$\text{vector } \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2s - t \\ s \\ 2t \\ t \end{bmatrix}$$

To see the underlying structure, we can separate the parameters s and t :

$$\mathbf{x} = \begin{bmatrix} -2s \\ s \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -t \\ 0 \\ 2t \\ t \end{bmatrix} = s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 2 \\ 1 \end{bmatrix}$$

This shows that every solution of the homogeneous system is a linear combination of the vectors $\mathbf{v}_1 = (-2, 1, 0, 0)$ and $\mathbf{v}_2 = (-1, 0, 2, 1)$, and conversely, every linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$ is a solution of the system. In other words, the solution set W is the set of all possible linear combinations of the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$, which is the same as the linear span of $\mathbf{v}_1$ and $\mathbf{v}_2$, i.e. $W = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$. So this solution set W is a subspace of $\mathbf{R}^4$. We also call this subspace W the solution space of $\mathbf{R}^4$.

In fact, the solution set of *any* homogeneous linear system with n variables will always be a subspace of $\mathbf{R}^n$.

To show why this is the case, let us consider a general homogeneous system $\mathbf{Ax} = \mathbf{0}$ (in matrix equation form) and check that its solution set W satisfies the three conditions of subspaces:

1. **W contain the zero vector:** The zero vector in $\mathbf{R}^n$ is $\mathbf{0} = (0, 0, \dots, 0)$. It is easy to see that the zero vector satisfies $\mathbf{Ax} = \mathbf{0}$ (by substituting $x = 0$). So, the zero vector is indeed a solution in W .
2. **W is closed under addition:** Let $\mathbf{u}$ and $\mathbf{v}$ be two arbitrary vectors in the solution set W . This means they are both solutions satisfying the system $\mathbf{Ax} = \mathbf{0}$, so $\mathbf{Au} = \mathbf{0}$ and $\mathbf{Av} = \mathbf{0}$. We need to check if their sum, $\mathbf{u} + \mathbf{v}$, is also in W :

$$\mathbf{A}(\mathbf{u} + \mathbf{v}) = \mathbf{Au} + \mathbf{Av} = \mathbf{0} + \mathbf{0} = \mathbf{0}$$
 Since $\mathbf{A}(\mathbf{u} + \mathbf{v}) = \mathbf{0}$, the vector $\mathbf{u} + \mathbf{v}$ is also a solution of $\mathbf{Ax} = \mathbf{0}$ and is therefore in W . So the set is closed under addition.
3. **W is closed under scalar multiplication:** Let $\mathbf{u}$ be a vector in W (so $\mathbf{Au} = \mathbf{0}$) and let c be any scalar. We need to check if the vector $c\mathbf{u}$ is also in W .

$$\mathbf{A}(c\mathbf{u}) = c(\mathbf{Au}) = c(\mathbf{0}) = \mathbf{0}$$
 Since $\mathbf{A}(c\mathbf{u}) = \mathbf{0}$, the vector $c\mathbf{u}$ is also a solution of $\mathbf{Ax} = \mathbf{0}$ and is therefore in W . So the set is closed under scalar multiplication.

Because the solution set W of our homogeneous system satisfies all three conditions, it is a subspace of $\mathbf{R}^n$.

Scenario 5

Let S be the solution set of homogeneous system

$$\begin{aligned}x_1 + 2x_2 - x_3 + 3x_4 &= 1 \\ 2x_1 + 4x_2 + x_3 + 0x_4 &= 2\end{aligned}$$

As the system has 4 variables (x_1, x_2, x_3 , and x_4), this solution set S can be regarded as a subset of $\mathbf{R}^4$.

The augmented matrix for this system is: $\begin{bmatrix} 1 & 2 & -1 & 3 & 1 \\ 2 & 4 & 1 & 0 & 2 \end{bmatrix}$

After performing Gauss Jordan elimination, we get the RREF $\begin{bmatrix} 1 & 2 & 0 & 1 & 1 \\ 0 & 0 & 1 & -2 & 0 \end{bmatrix}$.

The columns of variables x_2 and x_4 are non-pivot columns, so we set them as parameters $x_2 = s$ and $x_4 = t$.

After expressing x_1 and x_3 in terms of the parameters s and t , we get the solution

$$\text{vector } \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 - 2s - t \\ s \\ 2t \\ t \end{bmatrix}$$

To see the underlying structure, we can separate the parameters s and t :

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -2s \\ s \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -t \\ 0 \\ 2t \\ t \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 2 \\ 1 \end{bmatrix}$$

While every solution of the system is a linear combination of the vectors $\mathbf{w} = (1, 0, 0, 0)$, $\mathbf{v}_1 = (-2, 1, 0, 0)$ and $\mathbf{v}_2 = (-1, 0, 2, 1)$, not every linear combination of $\mathbf{w}$, $\mathbf{v}_1$ and $\mathbf{v}_2$ is a solution of the system. (Note that we do not have a parameter as the scalar for the term $\mathbf{w} = (1, 0, 0, 0)$.) So the solution set S is not the same as the linear span of $\mathbf{w}$, $\mathbf{v}_1$ and $\mathbf{v}_2$. i.e. $W = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$. In fact, this solution set S is not a subspace of $\mathbf{R}^4$.

It is easy to see that the solution set S of *any* non-homogeneous linear system $\mathbf{Ax} = \mathbf{b}$ (with n variables) can never be a subspace of $\mathbf{R}^n$, as zero vector $\mathbf{0} = (0, 0, \dots, 0)$ does not satisfy $\mathbf{Ax} = \mathbf{b}$ when $\mathbf{b}$ is non-zero. So S does not contain the zero vector. Since it violates the first condition of a subspace, S is not a subspace of $\mathbf{R}^n$. We can check that S also violates the other two conditions on the closure properties under vector addition and scalar multiplication.